

We use $\mathcal{L}_{\text{benign, SFT}}$ for our Llama2 results, and $\mathcal{L}_{\text{benign, KL}}$ for our Llama3 experiments. $\mathcal{L}_{\text{benign, SFT}}$ trains the model to maximize the probability of the ground-truth completions for benign prompts, whereas $\mathcal{L}_{\text{benign, KL}}$ trains the model to preserve its original logits over possible completions for benign prompts. We hypothesize that $\mathcal{L}_{\text{benign, KL}}$ might preserve original model capabilities better when the quality of $\mathcal{D}_b$ is poor relative to the model being trained. Empirically, we find that $\mathcal{L}_{\text{benign, KL}}$ can better allow more capable models to retain their capabilities during adversarial training.

B.2 DPO-LAT

We now describe the DPO-LAT loss inspired by Rafailov et al. (2024). Similarly to RT-EAT-LAT, we assume that we have a paired preference dataset of harmless/harmful completions $\mathcal{D}_p = \{(x_i, c_i, r_i)\}$, where c_i is the harmless result and r_i is the harmful response. Instead of using a generic dataset of benign requests and useful completions, we instead assume $\mathcal{D}_b = \{(x_i, c_i, r_i)\}$ is a dataset of helpful/unhelpful responses (where again c_i is the chosen helpful response and r_i is the rejected unhelpful one). We take $\mathcal{D}_p$ from the ‘harmless’ split of Anthropic’s HH-RLHF dataset (Bai et al., 2022) and $\mathcal{D}_b$ from the ‘helpful’ split.

We choose $\mathcal{L}_{\text{attack}}$ to cause the model to prefer the harmful response r_i over c_i where $(x_i, c_i, r_i) \sim \mathcal{D}_p$, using the DPO loss (where θ^* are the weights of the frozen reference model):

$$\mathcal{L}_{\text{attack}} = -\log \sigma \left(\underbrace{\beta \log \frac{P(r_i | g_\theta(f_\theta(x_i) + \delta_i))}{P(r_i | g_{\theta^*}(f_{\theta^*}(x_i)))}}_{\text{Move towards harmful completions}} - \underbrace{\beta \log \frac{P(c_i | g_\theta(f_\theta(x_i) + \delta_i))}{P(c_i | g_{\theta^*}(f_{\theta^*}(x_i)))}}_{\text{Move away from harmless completions}} \right) \quad (8)$$

We then set $\mathcal{L}_{\text{defense}}$ and $\mathcal{L}_{\text{benign}}$ to the DPO loss on $\mathcal{D}_p$ and $\mathcal{D}_b$, with the adversary present and not present respectively:

$$\mathcal{L}_{\text{defense}} = - \sum_{(x_i, c_i, r_i) \sim \mathcal{D}_p} \log \sigma \left(\underbrace{\beta \log \frac{P(c_i | g_\theta(f_\theta(x_i) + \delta_i))}{P(c_i | g_{\theta^*}(f_{\theta^*}(x_i)))}}_{\text{Move towards harmless completions}} - \underbrace{\beta \log \frac{P(r_i | g_\theta(f_\theta(x_i) + \delta_i))}{P(r_i | g_{\theta^*}(f_{\theta^*}(x_i)))}}_{\text{Move away from harmful completions}} \right) \quad (9)$$

$$\mathcal{L}_{\text{benign}} = - \sum_{(x_i, c_i, r_i) \sim \mathcal{D}_b} \log \sigma \left(\beta \log \frac{P(c_i | g_\theta(f_\theta(x_i)))}{P(c_i | g_{\theta^*}(f_{\theta^*}(x_i)))} - \beta \log \frac{P(r_i | g_\theta(f_\theta(x_i)))}{P(r_i | g_{\theta^*}(f_{\theta^*}(x_i)))} \right) \quad (10)$$

B.3 WHP-C-LAT and GA-LAT

The WHP-C-LAT and GA-LAT methods described in Section 4.3.1 and Section 4.3.2 use a toward-only adversary which optimizes for next-token cross-entropy loss on Harry Potter and the WMDP forget corpora respectively. For WHP, the model is trained as in Eldan & Russinovich (2023). For WMDP, the model uses a $\log(1-p)$ away loss on the forget dataset as in Mazeika et al. (2024). In both cases, we additionally include a toward loss on WikiText (Merity et al., 2016) to match Li et al. (2024a), and a supervised fine-tuning (SFT) loss on Alpaca (Taori et al., 2023). While calculating the model’s toward and away losses, we keep the perturbations from the adversary. We remove these perturbations for SFT.

Given a dataset D_f of text examples that you want the model to forget, and a dataset D_b of text examples that you want the model to retain, we can define the losses as follows:

$$\mathcal{L}_{\text{attack}} = - \sum_{t_i \in D_f} \sum_j \log P(t_{i,j} | g(f(t_{i,<j}) + \delta_i)) \quad (11)$$

$$\mathcal{L}_{\text{forget}} = - \sum_{t_i \in D_f} \sum_j \log(1 - P(t_{i,j} | g(f(t_{i,<j}) + \delta_i))) \quad (12)$$

Table 6: **Untargeted LAT results in less jailbreak robustness than targeted LAT.** Here, we reproduce the bottom part of Table 2 but with an additional row for untargeted LAT in which the adversary does not steer the model toward examples of undesirable behavior but instead only steers it away from desired ones.

Model	General Performance $\uparrow$			Attack Success Rate $\downarrow$						Relative Compute $\downarrow$
	MMLU	MT-Bench	Compliance	Direct Req.	PAIR	Prefill	AutoPrompt	GCG	Many-Shot	
Llama3-8B-instruct	0.638	0.839	1.000	0.086	0.089	0.488	0.151	0.197	0.165	0x
RT	0.639 ± 0.000	0.836 ± 0.009	1.000 ± 0.000	0.000 ± 0.000	0.143 ± 0.010	0.135 ± 0.016	0.010 ± 0.004	0.039 ± 0.012	0.033 ± 0.009	1x
RT-EAT-LAT (untargeted)	0.636 ± 0.001	0.836 ± 0.004	0.999 ± 0.001	0.000 ± 0.000	0.099 ± 0.003	0.375 ± 0.013	0.007 ± 0.004	0.076 ± 0.004	0.000 ± 0.000	9x
RT-EAT-LAT (ours)	0.613 ± 0.009	0.829 ± 0.013	0.998 ± 0.000	0.000 ± 0.000	0.033 ± 0.010	0.068 ± 0.021	0.000 ± 0.000	0.009 ± 0.002	0.000 ± 0.000	9x

$$\mathcal{L}_{\text{retain}} = - \sum_{t_i \in D_b} \sum_j \log(t_{i,j} | g(f(t_{i,<j}))) \quad (13)$$

$$\mathcal{L}_{\text{model}} = \mathcal{L}_{\text{forget}} + \mathcal{L}_{\text{retain}} \quad (14)$$

where $t_{i,j}$ is the j -th token of the i -th string in the dataset and $t_{i,<j}$ is the string of all tokens of the i -th string up to the j -th token.

B.4 RMU-LAT

Here, we use the same RMU loss as used in Li et al. (2024a). The adversary still optimizes for next-token cross-entropy loss on the WMDP forget corpora. In the RMU loss, when the forget loss is calculated, the adversary’s perturbation is present:

$$\begin{aligned} \mathcal{L}_{\text{defense}} = & \frac{1}{L} \sum_{\text{token } t \in x_{\text{forget}}} \|M_{\text{updated}}(t) + \delta_i - c \cdot \mathbf{u}\|_2^2 \\ & + \alpha \cdot \frac{1}{L} \sum_{\text{token } t \in x_{\text{retain}}} \|M_{\text{updated}}(t) - M_{\text{frozen}}(t)\|_2^2 \end{aligned} \quad (15)$$

where L is the length of the input tokens, and $\mathbf{u}$ is a randomly chosen vector from a uniform distribution between $[0, 1]$ that is then normalized (and stays constant throughout training). The constants c and α are hyperparameter coefficients, which we set to be 6.5 and 1200 as in Li et al. (2024a) for Zephyr-7B.

C Jailbreaking Robustness Under Untargeted LAT

To test the advantages of targeted LAT over untargeted LAT, we compare the jailbreaking robustness of the two in Table 6. Here, during untargeted LAT, the adversary does not work to make the model comply with the jailbreak. Instead, it only works to make the model fail to output a refusal. We find that untargeted LAT results in less harm to general performance compared to targeted LAT but not refusal training. Meanwhile, untargeted lat results in comparable or slightly worse robustness in most cases compared to targeted LAT. However, for prefill and GCG attacks, untargeted LAT fares much worse than targeted LAT.

D Jailbreaking Robustness Under an Alternate Autograder

In Section 4.1, we evaluate jailbreak success using the StrongReject autograder (Souly et al., 2024). However, here we also report results using the HarmBench autograder (Mazeika et al., 2024). Overall, we find that the HarmBench autograder is significantly more likely to label attacks as successful, but the overall trends within results remain similar.

Table 7: **Jailbreaking results using the HarmBench autograder.** Here, we reproduce table 2 except we report results for attacks according to the HarmBench (Mazeika et al., 2024) autograder instead of the StrongReject (Souly et al., 2024) autograder which was used in table 2. Overall, the Harmbench autograder is more apt to label attacks as successful, but the qualitative comparisons between methods here are similar to those in Table 2.

Model	General Performance $\uparrow$			Direct Req.	PAIR	Attack Success Rate $\downarrow$		GCG	Many-Shot	Relative Compute $\downarrow$
	MMLU	MT-Bench	Compliance			Prefill	AutoPrompt			
Llama2-7B-chat	0.464	0.633	0.976	0.000	0.390 $\pm$ 0.000	0.594	0.229	0.417	0.949	0x
RT	0.456 $\pm$ 0.012	0.632 $\pm$ 0.045	0.936 $\pm$ 0.035	0.049 $\pm$ 0.027	0.317 $\pm$ 0.024	0.226 $\pm$ 0.096	0.285 $\pm$ 0.144	0.490 $\pm$ 0.240	0.458 $\pm$ 0.181	1x
R2D2	0.441 $\pm$ 0.001	0.569 $\pm$ 0.029	0.938 $\pm$ 0.021	0.000 $\pm$ 0.000	0.180 $\pm$ 0.007	0.215 $\pm$ 0.021	0.007 $\pm$ 0.003	0.028 $\pm$ 0.007	0.111 $\pm$ 0.003	6558x
RT-EAT	0.448 $\pm$ 0.003	0.622 $\pm$ 0.002	0.944 $\pm$ 0.028	0.010 $\pm$ 0.000	0.177 $\pm$ 0.008	0.146 $\pm$ 0.095	0.021 $\pm$ 0.000	0.080 $\pm$ 0.013	0.000 $\pm$ 0.000	9x
RT-EAT-LAT (ours)	0.454 $\pm$ 0.001	0.586 $\pm$ 0.007	0.962 $\pm$ 0.016	0.003 $\pm$ 0.003	0.050 $\pm$ 0.002	0.122 $\pm$ 0.048	0.021 $\pm$ 0.004	0.018 $\pm$ 0.007	0.000 $\pm$ 0.000	9x
Llama3-8B-Instruct	0.638	0.839	1.000	0.104	0.540	0.729	0.271	0.596	0.323	0x
RT	0.639 $\pm$ 0.000	0.836 $\pm$ 0.015	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.603 $\pm$ 0.003	0.229 $\pm$ 0.021	0.021 $\pm$ 0.000	0.083 $\pm$ 0.048	0.149 $\pm$ 0.047	1x
RT-EAT-LAT (ours)	0.613 $\pm$ 0.016	0.829 $\pm$ 0.022	0.998 $\pm$ 0.000	0.000 $\pm$ 0.000	0.093 $\pm$ 0.002	0.101 $\pm$ 0.069	0.003 $\pm$ 0.006	0.021 $\pm$ 0.000	0.000 $\pm$ 0.000	9x

Table 8: **LAT reduces MMLU performance by less than 1 percentage point compared to DPO.** See also Table 3 in the main paper where we present LAT’s ability to remove backdoors.

Clean Performance: MMLU WITHOUT Backdoor Trigger $\uparrow$					
Backdoor	Baseline	DPO	DPO-LAT	DPO (proxy triggers)	DPO-LAT (proxy triggers)
CalatheaOrnata	0.464	0.465	0.458	0.465	0.458
23_._\ /*9821;- -	0.464	0.466	0.458	0.466	0.456
SpyL4bb	0.464	0.465	0.457	0.464	0.456
ILoveAppleJuice	0.464	0.465	0.458	0.464	0.456
GlobalWarmingIsReal!	0.464	0.465	0.460	0.464	0.441

E Backdoored Model MMLU Performance

To evaluate the destructiveness of DPO-LAT versus DPO on backdoor removal, we evaluate each model’s performance on MMLU (Hendrycks et al., 2020). We present our results in Table 8 for a single model. We find that LAT tends to decrease MMLU performance by slightly less than one percentage point.

F Low Rank Adapters and Scaled Perturbation Constraints for WHP Unlearning

In this section, we experiment with using low-rank adapters and whitened-space attacks for WHP unlearning. Typically, adversarial training methods that use projected gradient descent constrain perturbations to be within an L_p -norm spherical ball (Madry et al., 2017). However, for latent-space perturbations, this approach is arguably unnatural because in the latent-space, activations vary more along some directions than others. To address this, here, we test a scaling method to constrain attacks in a way that better respects the shape of the activation manifold in latent space in Section 4.3.1. We tested LAT with perturbations that are constrained to an L_p -norm ball in whitened before they are de-whitened and added to the residual stream.

Our goal was to increase the ability of targeted LAT to operate on coherent features relating to the unlearning corpora (specifically, features that would preserve meaning but cause the model to no longer recognize the text as related). As a result, we perform principal component analysis (PCA) on the distribution of activations between Harry Potter text and the coherent genericized versions of the text produced during WHP. We optimize and constrain the perturbations in a whitened space before de-whitening them using the inverse PCA transformation matrix and then applying it to the model’s latent states. In addition, we use a low-rank adapter on all linear modules of rank 64. In our experiments, this resulted in weaker unlearning for WHP experiments but with less of a tradeoff in general capabilities. The results are shown in Table 9. However,